

Summer term assessment

Answer every question. Show methods and explanations. Each question is worth 2 marks.

Name: _____ Date: _____

1. [2] Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.

6. [2] Prove or explain your reasoning: Convert 2.65 kg to grams and 3,750 ml to litres.

2. [2] Prove or explain your reasoning: Find the missing terms: 84, __, 62, __, 40.

7. [2] Find the area of a triangle with base 15 cm and perpendicular height 8 cm.

3. [2] Solve $6(x + 4) = 72$.

8. [2] Prove or explain your reasoning: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.

4. [2] Prove or explain your reasoning: Solve $5x - 7 = 48$.

9. [2] A cuboid has volume 360 cm³, length 12 cm and width 5 cm. Find its height.

5. [2] Write 845 mm in metres.

10. [2] Prove or explain your reasoning: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

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11. [2] A triangle has angles 47 degrees and 68 degrees. Find the third angle.

16. [2] Prove or explain your reasoning: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

12. [2] Prove or explain your reasoning: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

17. [2] A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?

13. [2] Translate (4, -2) by 3 left and 5 up.

18. [2] Prove or explain your reasoning: Three fifths of a number is 84. Find 35% of the number.

14. [2] Prove or explain your reasoning: Point P moves from (2, -5) to (-3, 1). Describe the translation.

19. [2] Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.

15. [2] In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.

20. [2] Prove or explain your reasoning: Solve $5x + 8 = 73$ and convert 3.45 km to metres.

Answers and evidence

1. [2] Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.
Answer $C = 6n + 4$.
2. [2] Prove or explain your reasoning: Find the missing terms: 84, __, 62, __, 40.
Answer 73 and 51. The explanation must show why the method works.
3. [2] Solve $6(x + 4) = 72$.
Answer $x = 8$.
4. [2] Prove or explain your reasoning: Solve $5x - 7 = 48$.
Answer $x = 11$. The explanation must show why the method works.
5. [2] Write 845 mm in metres.
Answer 0.845 m.
6. [2] Prove or explain your reasoning: Convert 2.65 kg to grams and 3,750 ml to litres.
Answer 2,650 g and 3.75 litres. The explanation must show why the method works.
7. [2] Find the area of a triangle with base 15 cm and perpendicular height 8 cm.
Answer 60 cm².
8. [2] Prove or explain your reasoning: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.
Answer 84 cm². The explanation must show why the method works.
9. [2] A cuboid has volume 360 cm³, length 12 cm and width 5 cm. Find its height.
Answer 6 cm.
10. [2] Prove or explain your reasoning: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
Answer 420 cm³. The explanation must show why the method works.

Answers and evidence

11. [2] A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer 65 degrees.
12. [2] Prove or explain your reasoning: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
Answer 68 degrees. The explanation must show why the method works.
13. [2] Translate (4, -2) by 3 left and 5 up.
Answer (1, 3).
14. [2] Prove or explain your reasoning: Point P moves from (2, -5) to (-3, 1). Describe the translation.
Answer 5 left and 6 up. The explanation must show why the method works.
15. [2] In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.
Answer 90 degrees.
16. [2] Prove or explain your reasoning: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?
Answer 60 pupils. The explanation must show why the method works.
17. [2] A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?
Answer 320 cm³.
18. [2] Prove or explain your reasoning: Three fifths of a number is 84. Find 35% of the number.
Answer 49. The explanation must show why the method works.
19. [2] Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.
Answer 99 cm² and 20.
20. [2] Prove or explain your reasoning: Solve $5x + 8 = 73$ and convert 3.45 km to metres.
Answer $x = 13$ and 3,450 m. The explanation must show why the method works.

Score bands and intervention response

Lower 0-15 / 40

Immediate structured intervention. Re-teach prerequisite concepts with concrete and visual representations.

Cuspy 16-23 / 40

Targeted short intervention. Address two or three misconceptions, then reassess within two weeks.

Expected 24-31 / 40

Secure core understanding. Revisit any domain below half marks and continue mixed retrieval.

Higher 32-40 / 40

Strong understanding. Use proof, generalisation and unfamiliar multi-step problems for depth.

Teacher decision

Use the band as a starting point. Review which questions were secure, classroom evidence, attendance, language needs and reasonable adjustments. Record a precise intervention, named adult, frequency and review date in the editable evaluation record.